

HTML Elements Guide

This guide organizes HTML elements into categories, building on the provided structure. Each category includes a list of relevant elements, followed by a table detailing the element, its uses, an example, and practical uses. Categories are expanded to cover all standard HTML elements (excluding obsolete/deprecated ones for modernity). The structure draws from standard HTML references for completeness.

Semantic Elements: For Structuring and Partitions

These elements provide meaning to the document structure, helping with accessibility, SEO, and layout without implying specific styling.

Elements:

- 1. div
- 2. header
- 3. nav
- 4. aside
- 5. footer
- 6. article
- 7. main
- 8. section
- 9. address
- 10. hgroup
- 11. search

Element	Uses	Example	Practical Uses
div	A generic container for grouping content, often used for styling or scripting purposes.	<code><div class="container">Content here</div></code>	Wrapping sections of a page for CSS layouts, like creating columns or modals in web apps.
header	Represents introductory content or navigational aids at the top of a page or section.	<code><header><h1>Site Title</h1></header></code>	Site banners, logos, and menus in websites; section intros in articles.
nav	Defines a set of navigation links.	<code><nav>Home</nav></code>	Main menus, sidebars, or pagination in blogs and e-commerce sites.
aside	Represents content indirectly related to the main content, like sidebars.	<code><aside><h3>Related Links</h3></aside></code>	Sidebars with ads, related articles, or author bios in news sites.
footer	Defines a footer for a document or section, typically containing copyright or contact info.	<code><footer>&copy; 2025 Company</footer></code>	Bottom sections of pages with links, social media, or disclaimers in corporate websites.
article	Represents self-contained content that could be distributed independently.	<code><article><h2>News Story</h2><p>Details...</p></article></code>	Blog posts, forum threads, or news articles in content-heavy sites.
main	Specifies the main content of the document, unique to the page.	<code><main><article>Primary content</article></main></code>	Central area of a webpage, excluding headers/footers, for better accessibility in apps.
section	Represents a standalone section of content, often with a heading.	<code><section><h2>Chapter 1</h2><p>Text...</p></section></code>	Dividing long pages into thematic parts, like in tutorials or reports.
address	Provides contact information for the author or owner of a	<code><address>Email: info@example.com</address></code>	Contact details in footers or about pages for businesses.

Element	Uses	Example	Practical Uses
	document/section.		
hgroup	Groups a set of heading elements (h1-h6) when the heading has multiple levels.	<hgroup> Main Title Subtitle </hgroup>	Complex headings in articles or books, though rarely used due to modern alternatives.
search	Represents a part of the page for search functionality.	<search><form><input type="search"></form></search>	Search bars in headers for e-commerce or wiki sites.

Content Centric Elements

These focus on presenting and organizing content, grouped into subcategories as suggested.

1. Text Elements

Elements for handling text, including headings, paragraphs, and inline formatting.

Elements: h1, h2, h3, h4, h5, h6, p, b, strong, em, i, u, s, small, sub, sup, abbr, code, samp, kbd, var, dfn, mark, q, cite, time, data, br, wbr, span, bdi, bdo, pre, blockquote, figcaption, figure

Element	Uses	Example	Practical Uses
h1 to h6	Define headings with decreasing importance (h1 most important).	Main Heading	Page titles (h1), subsections (h2-h6) for SEO and structure in documents.
p	Defines a paragraph of text.	This is a paragraph.	Body text in articles, blogs, or descriptions.
b	Bold text without added importance.	Bold text	Styling keywords in UI, though strong is preferred for semantics.
strong	Indicates strong importance, typically bold.	Important!	Emphasizing warnings or key terms in instructions.
em	Emphasized text, typically italic.	Emphasized	Stressing words in sentences for readability in narratives.
i	Italic text without emphasis.	<i>Italic</i>	Foreign words or thoughts in text.
u	Underlined text.	<u>Underlined</u>	Misspelled words in editors or links (though a is better for links).
s	Strikethrough text for deleted content.	<s>Old price: \$10</s>	Showing discounts in e-commerce or edits in wikis.
small	Smaller text for side comments.	Copyright notice	Fine print in footers or disclaimers.
sub	Subscript text.	H₂O	Chemical formulas or footnotes.
sup	Superscript text.	x²	Math equations or ordinal numbers like 1 st .

Element	Uses	Example	Practical Uses
abbr	Abbreviation with optional title for expansion.	<abbr title="HyperText Markup Language">HTML</abbr>	Tooltips for acronyms in technical docs.
code	Inline code snippet.	<code>console.log()</code>	Displaying code in tutorials.
samp	Sample output from a program.	<samp>Error: 404</samp>	Showing console outputs in docs.
kbd	Keyboard input.	<kbd>Ctrl + C</kbd>	Instructions for shortcuts in software guides.
var	Variable in math or programming.	<var>x</var>	Math formulas or code explanations.
dfn	Defining instance of a term.	<dfn>HTML</dfn> is a markup language.	Glossaries or first mentions in articles.
mark	Highlighted text for reference.	<mark>Highlighted</mark>	Search results or annotations.
q	Inline quotation.	<q>Quote here</q>	Short quotes in text.
cite	Citation of a creative work.	<cite>Book Title</cite>	Referencing sources in blogs.
time	Machine-readable date/time.	<time datetime="2025-08-18">Today</time>	Event dates for calendars or SEO.
data	Machine-readable data.	<data value="123">Item</data>	Linking data for scripting in lists.
br	Line break.	Line one Line two	Poetry or addresses without paragraphs.
wbr	Word break opportunity.	Verylong<wbr>word	Preventing awkward line breaks in long words.
span	Generic inline container.	Text	Styling parts of text without semantics.
bdi	Isolates text for bidirectional control.	<bdi>Arabic text</bdi>	Mixing LTR/RTL languages.
bdo	Overrides text direction.	<bdo dir="rtl">Text</bdo>	Forcing direction in multilingual sites.
pre	Preformatted text, preserves whitespace.	<pre>Code block</pre>	Displaying code or ASCII art.
blockquote	Block quotation.	<blockquote>Long quote</blockquote>	Pull quotes in articles.
figcaption	Caption for a figure.	<figure><figcaption>Caption</figcaption></figure>	Image descriptions in galleries.
figure	Self-contained content like images with caption.	<figure></figure>	Illustrations or charts in reports.

2. Lists

Elements for creating ordered, unordered, or description lists.

Elements: ol, ul, li, dl, dt, dd, menu

Element	Uses	Example	Practical Uses
ol	Ordered list.	Item 1	Numbered steps in recipes or tutorials.
ul	Unordered list.	Item	Bullet points in features lists.
li	List item in ol/ul/menu.	Item	Individual entries in menus or checklists.
dl	Description list.	<dl><dt>Term</dt><dd>Definition</dd></dl>	Glossaries or FAQs.
dt	Term in description list.	<dt>HTML</dt>	Keys in key-value pairs.
dd	Description in description list.	<dd>Markup language</dd>	Values in definitions.
menu	List of commands.	<menu> Option </menu>	Context menus, though rarely used.

3. Tables

Elements for tabular data.

Elements: table, caption, colgroup, col, thead, tbody, tfoot, tr, th, td

Element	Uses	Example	Practical Uses
table	Defines a table.	<table><tr><td>Cell</td></tr></table>	Data grids like spreadsheets or schedules.
caption	Table title.	<caption>Table Title</caption>	Describing table content for accessibility.
colgroup	Groups columns for styling.	<colgroup><col span="2"></colgroup>	Applying CSS to column groups.
col	Defines column properties.	<col style="width:50%">	Setting widths in tables.
thead	Table header section.	<thead><tr><th>Header</th></tr></thead>	Column headers in reports.
tbody	Table body section.	<tbody><tr><td>Data</td></tr></tbody>	Main data rows.
tfoot	Table footer section.	<tfoot><tr><td>Total</td></tr></tfoot>	Summary rows like totals.
tr	Table row.	<tr><td>Cell</td></tr>	Rows in data tables.
th	Table header cell.	<th>Header</th>	Bold/centered headers.
td	Table data cell.	<td>Data</td>	Standard cells.

4. Media

Elements for embedding images, audio, video, and other media.

Elements: img, audio, video, source, track, picture, area, map, iframe, embed, object

Element	Uses	Example	Practical Uses
img	Embeds an image.		Photos in galleries or logos.
audio	Embeds sound content.	<audio src="sound.mp3" controls></audio>	Podcasts or background music.
video	Embeds video content.	<video src="video.mp4" controls></video>	Tutorials or ads on sites.
source	Specifies multiple media resources.	<video><source src="video.mp4" type="video/mp4"></video>	Fallbacks for different formats.

Element	Uses	Example	Practical Uses
track	Timed text track for media.	<video><track src="subtitles.vtt" kind="subtitles"></video>	Captions for accessibility.
picture	Responsive images with multiple sources.	<picture><source srcset="large.jpg" media="(min-width:600px)"></picture>	Adaptive images for devices.
area	Defines clickable areas in image maps.	<map><area shape="rect" coords="0,0,50,50" href="link"></map>	Interactive maps or diagrams.
map	Defines an image map.	<map name="map"></map>	Hotspots on images.
iframe	Embeds another HTML page.	<iframe src="page.html"></iframe>	Embedding maps or videos from external sources.
embed	Embeds external content like plugins.	<embed src="flash.swf">	Legacy plugins, rarely used now.
object	Embeds multimedia or resources.	<object data="pdf.pdf"></object>	PDFs or SVGs in pages.

5. Forms

Elements for user input and data submission.

Subcategories as provided:

Inputs

Elements/types: input (with types: text, email, password, number, color, submit, button, file, date, time, month, week, range, search, tel, url, hidden), button

Element/Type	Uses	Example	Practical Uses
input type="text"	Single-line text input.	<input type="text" name="name">	Name fields in sign-up forms.
input type="email"	Email address input with validation.	<input type="email" name="email">	Contact forms.
input type="password"	Masked text input.	<input type="password" name="pass">	Login screens.
input type="number"	Numeric input with spinner.	<input type="number" name="age">	Quantity selectors in shops.
input type="color"	Color picker.	<input type="color" name="color">	Theme customizers.
input type="submit"	Form submission button.	<input type="submit" value="Send">	Submit buttons in forms.
input type="button"	Generic button.	<input type="button" value="Click">	Triggering scripts.

Element/Type	Uses	Example	Practical Uses
input type="file"	File upload.	<input type="file" name="file">	Profile picture uploads.
input type="date"	Date picker.	<input type="date" name="dob">	Booking calendars.
input type="time"	Time picker.	<input type="time" name="time">	Appointment scheduling.
input type="month"	Month/year picker.	<input type="month" name="month">	Credit card expiration.
input type="week"	Week picker.	<input type="week" name="week">	Weekly reports.
input type="range"	Slider for numeric range.	<input type="range" min="0" max="100">	Volume controls.
input type="search"	Search input with clear button.	<input type="search" name="search">	Site search bars.
input type="tel"	Telephone number input.	<input type="tel" name="phone">	Contact info.
input type="url"	URL input with validation.	<input type="url" name="website">	Link submissions.
input type="hidden"	Hidden data sent with form.	<input type="hidden" name="id" value="123">	Session tokens.
button	Clickable button in forms.	<button type="submit">Submit</button>	Styled submit buttons.

Selection

Elements: select, option, optgroup, datalist

Element	Uses	Example	Practical Uses
select	Dropdown list.	<select><option>Value</option></select>	Country selectors in forms.
option	Item in select or datalist.	<option value="1">Option 1</option>	Choices in menus.
optgroup	Groups options in select.	<optgroup label="Group"><option>Opt</option></optgroup>	Categorized dropdowns like continents/countries.
datalist	Predefined options for input.	<input list="options"><datalist id="options"><option>Value</option></datalist>	Autocomplete suggestions.

Radio

Element: input type="radio"

Element/Type	Uses	Example	Practical Uses
input type="radio"	Single choice from group.	<input type="radio" name="group" value="1">	Gender selection or polls.

Check

Element: input type="checkbox"

Element/Type	Uses	Example	Practical Uses
input type="checkbox"	Multiple choices.	<input type="checkbox" name="option">	Terms agreement or multi-select filters.

Additional Form Elements: form, label, fieldset, legend, textarea, output, progress, meter

Element	Uses	Example	Practical Uses
form	Container for form controls.	<form action="/submit"><input></form>	Collecting user data for login or surveys.
label	Caption for form control.	<label for="id">Label</label><input id="id">	Improving accessibility by linking to inputs.

Element	Uses	Example	Practical Uses
fieldset	Groups related form elements.	<fieldset><legend>Group</legend><input></fieldset>	Organizing long forms into sections.
legend	Caption for fieldset.	<legend>Section Title</legend>	Titles for grouped inputs.
textarea	Multi-line text input.	<textarea name="message"></textarea>	Comment boxes or descriptions.
output	Result of a calculation.	<output name="result">0</output>	Displaying form computations like totals.
progress	Progress of a task.	<progress value="50" max="100"></progress>	File upload indicators.
meter	Scalar measurement.	<meter value="0.6">60%</meter>	Battery levels or ratings.

Additional Categories for Completeness

Root and Metadata Elements

Foundational elements for document structure.

Elements: html, head, body, base, link, meta, style, title

Element	Uses	Example	Practical Uses
html	Root element of HTML document.	<html lang="en"></html>	Enclosing all content; setting language.
head	Container for metadata.	<head><title>Page</title></head>	Holding scripts, styles, and meta info.
body	Main content of the document.	<body> Content </body>	All visible page content.
base	Base URL for relative links.	<base href="https://example.com/">	Simplifying links in multi-page sites.
link	External resource link, like CSS.	<link rel="stylesheet" href="style.css">	Loading stylesheets or favicons.
meta	Metadata like charset or viewport.	<meta charset="UTF-8">	SEO keywords, responsive design.
style	Inline CSS styles.	<style>body {color: red;}</style>	Page-specific styling.
title	Document title.	<title>Page Title</title>	Browser tab text and bookmarks.

Inline Edits and Other Semantics

For marking changes or special text.

Elements: del, ins, hr, ruby, rp, rt

Element	Uses	Example	Practical Uses
del	Deleted text.	Old text	Showing revisions in documents.
ins	Inserted text.	<ins>New text</ins>	Tracking changes in collaborative edits.
hr	Thematic break (horizontal rule).	<hr>	Separating sections in articles.
ruby	Ruby annotations for East Asian text.	<ruby>漢<rt>kan</rt></ruby>	Pronunciations in Japanese/Chinese sites.
rp	Fallback for ruby parentheses.	<ruby>漢<rp>(</rp><rt>kan</rt><rp>)</rp></ruby>	Compatibility for non-supporting browsers.
rt	Ruby text.	<rt>Annotation</rt>	Furigana in educational content.

Scripting and Interactive Elements

For dynamic content and user interaction.

Elements: script, noscript, canvas, details, summary, dialog

Element	Uses	Example	Practical Uses
script	Embeds executable code, usually JavaScript.	<script>alert('Hi');</script>	Adding interactivity like validations.
noscript	Content for when scripts are disabled.	<noscript>Enable JS</noscript>	Fallback messages in apps.
canvas	Drawable graphics via scripting.	<canvas width="200" height="100"></canvas>	Charts, games, or animations.
details	Disclosure widget for hide/show.	<details><summary>Click</summary> Hidden </details>	FAQs or expandable sections.
summary	Heading for details element.	<summary>Toggle</summary>	Accordion menus.
dialog	Modal or popup dialog.	<dialog open> Message </dialog>	Alerts or forms in web apps.

Web Components and Advanced

For custom elements and embeds.

Elements: template, slot, svg, math

Element	Uses	Example	Practical Uses
template	Holds client-side content not rendered initially.	<template> Content </template>	Reusable HTML in JavaScript frameworks.
slot	Placeholder in web components.	<slot name="content"></slot>	Custom elements in Shadow DOM.
svg	Container for SVG graphics.	<svg><circle cx="50" cy="50" r="40"/></svg>	Icons, charts, or illustrations.
math	MathML for mathematical notation.	<math><mi>x</mi></math>	Equations in scientific papers.